

defence SUBMARINE IMPASSE

The navy refuses to accept a missile-firing submarine refitted in Russia after it repeatedly fails test firings

DUD MISSILE

■ By Sandeep Unnithan

It was dubbed Baby Tomahawski, a short-legged Russian version of the eponymous US cruise missiles. Fired from the torpedo tube of a submarine, it could fly over 300 km delivering a 400 kg warhead to a target with pin-point accuracy. It would give the Indian Navy's submarines capability to attack targets on land. Yet, the Kilo class submarine *INS Sindhuvi* and the land-attack missiles it is equipped to fire have today joined aircraft carrier *Vikramaditya* as another bone of contention that India has with its largest defence supplier, Russia.

The navy recently refused to take the delivery of *Sindhuvi*, which completed refit at the Zvyozdochka shipyard near St Petersburg after its missiles failed to find their targets. The 50-man crew sent to Russia to sail back with the submarine is set to return to India with the naval brass taking a stern line by asking the shipyard to fix the problem before it takes the delivery of the submarine. The *Sindhuvi*, which completed its year-long refit in Russia last year, sailed to a test range in the Barents Sea for a series of firings on land targets. The 3M54E1 Klub cruise missiles, however, failed to hit the targets in as many as six consecutive test firings carried out in the Arctic between September and November. In certain cases, the snub-nosed subsonic missiles veered several kilometres away from their intended targets leading the navy to take this drastic step. It will take another year to rectify the defects and prove it in trials.

Delays seem to have become a common feature with some of the recently acquired Russian equipment. Last year, India had suspended payments for one of the three IL-38 maritime patrol aircraft upgraded with the Sea Dragon submarine detection equipment, because they did not meet performance parameters. The navy has detected hull vibrations on the three Talwar class frigates bought from Russia during 2003-04 and interface issues with the Ka-31 airborne early warning (AEW) helicopters flown off these frigates. Russia has also indicated a six-month delay in the delivery of the Akula-II nuclear-powered submarine, *Chakra*, moored at the far



eastern shipyard of Komsomolsk on Amur. The navy plans to lease it for 10 years to train crew to man its indigenous nuclear submarine, the Advanced Technology Vessel. The Kilo class submarine involves two variants of a missile, which arms eight of the navy's fleet of 10 Soviet-built Kilo class submarines acquired between 1985 and 2000. In the past five years, the submarines were sent to Russia for refits which involved not only refurbishment but also modification of their torpedo tubes to allow them to fire tube-launched missiles, giving the navy the capability to attack surface ships from submarines. So impressed was the navy with the Klub missile—with ranges in excess of 200 km they are twice the range of competing systems like the Harpoon—that it became its first customer. It ordered 175 missiles for its warships and submarines in 1998, receiving them between 2001 and 2006.

The *Sindhughosh* and the *Sindhuvi* were equipped with the new land-attack variant costing roughly Rs 30 crore each. These new missiles are crucial to influencing the outcome of a land battle by attacking targets like enemy airbases and infrastructure facilities. Two years ago, the *Sindhughosh* was equipped with land attack missile in Russia, becoming the first Indian submarine to be equipped thus. Coming out of its refit in Zvyozdochka shipyard it hit targets on land in the Barents Sea two times out of the four test firings. The 50 per cent hit rate was regarded as acceptable and the navy took delivery. In the past few years, how-

FROM RUSSIA WITH ANGST

Glitches and delayed deliveries cast a cloud over defence ties with Russia

IL-38 AIRCRAFT



Sea Dragon surface/submarine hunting system not

functioning in one of the three maritime patrol aircraft. Navy has halted payment and asked for the defect to be fixed.

AKULA-II N-SUBMARINE



Meant to train Indian crew to man the indigenous nuclear submarine, it is delayed by over 6 months. Officials suspect it could be a pressure tactic to get India to pay for *Vikramaditya*.

KLUB ANTI-SHIP MISSILE

India has bought over 100 of these missiles fitted on Kilo class submarines that were refitted by Russia. They have failed numerous test firings.



KA-31 HELICOPTER



The airborne early warning helicopter design has compatibility issues with the Talwar class frigates.